

1.

Step	The heuristic is to count how many letters differ from the goal. Each wrong letter needs at least one change, and each change costs 1 so you still need at least that many steps. It can not overestimate because the heuristic is always equal to the minimum amount of moves and you can only fix 1 letter per move.
Scrabble	For each letter that doesn't match the goal, take the sum of scrabble values to get the heuristic output. The cost it lower bounds is the cost of writing the missing goal letters. It can not overestimate because the goal letter placements are required. The heuristic also ignores extra wrong letters and extra moves only add cost . This ensures the cost is always $\geq$ current sum.
Frequency	The heuristic is to cumulatively add 1 + the rarity of the goal word. The unavoidable cost is the last move (1 + rarity(goal)). This cannot be overestimated because any complete path must include at least that final step.

2.

Part 1: Across all 6 experiments, UCS and A* returned the same optimal cost. WHY → ASK gave 5 for steps, 9 for scrabble, and 38.6 for frequency. BEAR → DUCK gave 4, 11, and 38.24.

Part 2: A* reduced expansions the most under scrabble for both pairs. WHY → ASK was reduced by 98.3%, and BEAR → DUCK was reduced by 99.8%. Steps had a similar sized reduction of 97.19% and 99.1% respectively.

Part 3: Frequency was the least informative. It had a smaller reductions as well of only 14.9% and 60.6%.

Part 4: Steps and scrabble increase with mismatched letters, so A* will prioritize better states. Frequency only adds 1 + rarity(goal) for every state. This gives less guidance so A* acts more like UCS in that case.

3.

	Cost symmetry?	Explanation
Step	Yes	Every legal move costs 1 in either direction. If a path from W1 to W2 has k steps, the reverse path also

		has k steps, so both will have the same total cost.
Scrabble	No	The cost is dependent on the exact letter that is inserted. The reverse path will insert different letters, leading to different costs. So the algorithm will flow in a different order.
Frequency	No	Frequency uses the goal word as part of determining the cost. In the reverse order, the goal word it uses will change, so the cost will generally differ.

4.

I used Gemini 3.8 Flash

Part 1:

Proposed Heuristic: $h_{\text{rarity}}(n)$

Let g be the goal word, and let $N(n) = \{w \in \text{Dict} \mid \text{Hamming}(n, w) = 1\}$ denote the set of legal one-letter neighbor words reachable from state n . Let $r_{\min} = \min_{w \in \text{Dict}} \text{rarity}(w) \geq 0$.

Part 2:

$$h(n) = \begin{cases} 0 & \text{if } n = g, \\ h_0(n) + \text{rarity}(g) & \text{if } h_0(n) = 1, \\ h_0(n) + \text{rarity}(g) + \min_{w \in N(n)} \text{rarity}(w) + (h_0(n) - 2) \cdot r_{\min} & \text{if } h_0(n) \geq 2. \end{cases}$$

(If $N(n) = \emptyset$ when $h_0(n) \geq 2$, state n is a dead end with infinite remaining cost; we can set $h(n) = \infty$.)

Part 3:

The heuristic seems admissible. You need at least h0 steps to reach the goal. If you're 2+ letters away, the first step must go to the cheapest neighbors rarity. Any other steps go through some dictionary. This is essentially just a more advanced version of the frequency heuristic.

Part 5:

Problem	Heuristic	Cost	Expanded
WHY → ASK	h0	38.86	755
WHY → ASK	Gemini	38.86	497
BEAR → DUCK	h0	38.24	720
BEAR → DUCK	Gemini	38.24	201

By recreating a function for this heuristic I was able to calculate the cost and nodes expanded similar to the other ones. This would support that Gemini's heuristic because it gives the same cost but expands less nodes. This means it is more informative.